

INFO4990 Honours Thesis Draft

Skill Representation for Scalable Agent Skill Retrieval under Semantic Confusability

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Abstract

Large language model agents increasingly rely on reusable skills: procedural capability units that may package instructions, workflow steps, scripts, examples, resources, and constraints. As skill libraries scale, the central difficulty is not only storing many capabilities, but selecting the right capability when multiple skills appear semantically similar. A request about extracting information from a document, for example, may plausibly match skills for document review, summarisation, citation grounding, field extraction, or format conversion, even though these skills require different procedures.

This thesis investigates what information agent skill representations must preserve to distinguish semantically similar but procedurally distinct skills at scale. It separates three layers that are often conflated in prior work: the skill artifact, the selection representation, and the retrieval policy. It then evaluates retrieval strategies that use flat metadata, full skill artifacts, structured procedural fields, and hybrid dense-retrieval plus reranking methods on a benchmark designed to stress semantic confusability. Preliminary experiments use a library of 1006 skills, including 67 controlled core skills and a large background skill set, with retrieval accuracy, top-k recall, ranking quality, context cost, latency, and failure modes as evaluation dimensions.

Current results suggest that description-only retrieval is fragile under scale, while full-skill representations and structured procedural fields improve robustness. The strongest current method combines dense retrieval over full skill artifacts with structure-aware schema reranking, supporting the claim that semantic retrieval and procedural representation are complementary rather than interchangeable. The expected contribution is a clearer framework for analyzing skill-selection systems, a controlled benchmark for procedurally confusable skill retrieval, and empirical evidence about which representational information matters for scalable agent skill libraries.

Declaration and Acknowledgements

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Chapter 1

Introduction

1.1 Background

Large language model (LLM) agents are increasingly designed as systems that combine reasoning, action, memory, and external capability use rather than as standalone text generators. Early work on retrieval-augmented generation showed that language models can improve performance by accessing external information at inference time rather than relying solely on knowledge stored in model parameters [1, 2]. Later work on tool use extended this idea from document retrieval to action execution by enabling models to call external functions and APIs [3, 4, 5].

More recently, agent systems have begun to treat reusable capabilities as *skills*. A skill is not simply an atomic tool call. It may include instructions, workflow steps, scripts, examples, dependencies, input requirements, output expectations, and constraints. In practical systems such as filesystem-based agent skill libraries, the full skill artifact may be stored as a directory containing a `SKILL.md` file plus optional resources. The agent may initially see only a compressed representation, such as a name and description, before deciding whether to load the full artifact.

This shift changes the retrieval problem. If the object being selected is a high-level procedure rather than an atomic function, retrieval quality depends not only on topical relevance, but also on procedural suitability. Two skills may look semantically similar because they refer to the same domain or task family, while still differing in preconditions, workflow steps, required resources, output format, or side effects.

1.2 Problem Statement

The problem addressed in this thesis is semantic confusion in large agent skill libraries. As skill libraries grow, a user request may match many plausible skills at the level of names and descriptions. The correct skill may exist, but the retrieval system may surface a nearby skill that sounds relevant while being procedurally wrong.

This failure mode is different from ordinary retrieval failure. The wrong result is often not unrelated. It is close enough to be tempting. For example, a request to “extract information from this PDF” may plausibly match document review, document summarisation, citation grounding, field extraction, or layout-preserving conversion. These skills may all mention documents, extraction, or PDFs, but they require different procedures and produce different outputs. A flat description may not preserve the procedural distinctions required for correct selection.

The scalability issue makes this worse. In a small library, the main agent may be able to inspect all skill names and short descriptions in context. In a larger library, even this metadata can become too large or noisy. The system then needs a candidate-subsetting stage before the main agent loads any full skill document. This makes the selection representation critical: the correct skill must be surfaced before the full artifact can be inspected.

1.3 Research Questions

This thesis uses the following working research questions:

RQ1: What information must agent skill representations preserve to distinguish semantically similar but procedurally distinct skills at scale?

RQ2: How do retrieval strategies that use flat text, dense embeddings, structured procedural fields, or hybrid reranking differ in accuracy, efficiency, and failure modes when applied to scalable skill libraries?

These questions revise an earlier framing that asked more generally how structure-aware representations improve over flat description-based retrieval. The revised framing keeps retrieval strategies in scope, but makes representation quality the central research object. The thesis therefore asks not only which method performs best, but what kinds of information are being preserved, lost, or exploited by each method.

1.4 Scope

The thesis focuses on retrieval and candidate selection before full skill execution. It does not attempt to build a complete general-purpose agent operating system. The main agent, the retriever, and the reranker are treated as separable components. The retriever may be a lexical method, an embedding model, a deterministic schema scorer, a neural reranker, or a hybrid pipeline. What matters is whether the retrieval stage can produce a candidate set that contains and ranks the procedurally correct skill.

The evaluation focuses on skill libraries where some skills are intentionally confusable. This is important because broad relevance retrieval is too easy for the research question. A benchmark that only asks whether a document task retrieves a document skill would not test the central failure mode. The difficult case is selecting between skills that share the same broad domain but

differ in operational requirements.

1.5 Contributions

This thesis aims to make four contributions:

1. A conceptual framework that separates skill artifacts, selection representations, and retrieval policies.
2. A benchmark design for scalable skill retrieval under semantic confusability and procedural distinctness.
3. An empirical comparison of flat metadata, full skill artifacts, structured procedural representations, and hybrid retrieval-reranking methods.
4. A failure-mode analysis identifying what information is needed for robust skill selection, including inputs, outputs, preconditions, workflow constraints, dependencies, resources, negative boundaries, and meta-level task roles.

1.6 Thesis Structure

Chapter 2 reviews the literature on retrieval-augmented language models, tool use, agent skills, skill-library scaling, and structure-aware retrieval. Chapter 3 defines the conceptual framework used throughout the thesis. Chapter 4 describes the benchmark design. Chapter 5 explains the methodology and evaluation metrics. Chapter 6 reports current and planned results. Chapter 7 discusses implications and failure modes. Chapter 8 concludes the thesis and outlines future work.

Chapter 2

Literature Review

Version Note

This chapter preserves the Assignment 3 literature review as the base thesis literature review. The research questions at the end of the original review are retained as historical wording from the assignment. The current working research questions are stated in Chapter 1 and should supersede the older wording when the final thesis is revised.

2.1 Introduction

Large language model (LLM) agents are increasingly designed as systems that combine reasoning, action, memory, and external capability use rather than as standalone text generators. Early work on retrieval-augmented generation showed that language models can improve performance by accessing external non-parametric memory at inference time rather than relying solely on knowledge stored in model parameters [1, 2]. Later work on tool use extended this idea from document retrieval to action execution by enabling models to call external functions and APIs [3, 4, 5]. More recently, the literature has begun to treat agent skills as a distinct abstraction: reusable procedural capabilities that may package instructions, scripts, workflows, tools, and task-specific resources into units that can be invoked across tasks [6, 7, 8]. Unlike atomic tools, which typically expose a specific callable function, skills operate as higher-level procedural modules. This shift is important because it changes the retrieval problem. If the objects being selected are not atomic tools but higher-level procedures, then retrieval quality depends not only on semantic relevance but also on procedural suitability.

The research problem addressed in this review is therefore not simply whether large capability libraries are difficult for agents to manage. That broad point is already well established. The more specific issue is whether current methods can distinguish between skills that appear semantically similar at the level of names or descriptions, yet differ in preconditions, workflow steps, dependencies, outputs, or side effects. This distinction matters particularly in personal-

agent settings, where users may accumulate many overlapping skills over time without maintaining a carefully curated capability library. In such settings, flat skill descriptions may create ambiguity even when the correct skill exists in the library. A retrieval system may therefore fail not because a capability is missing, but because the representation and selection process do not preserve procedural differences strongly enough.

This literature review is organised around that problem. Rather than summarising papers one by one, it examines the literature through the questions most relevant to scalable skill retrieval. First, why should skills be treated as a distinct retrieval problem rather than as a simple extension of tool selection? Second, what does the emerging skill literature reveal about the scaling problem itself? Third, what representations and retrieval mechanisms have been proposed for skills, and what limitations remain? Finally, what evaluation practices are currently used, and do they adequately test semantic confusability, context efficiency, and downstream task performance? The review therefore compares prior work not only in terms of whether it improves overall agent behaviour, but also in terms of what kind of skill representation it assumes, how retrieval is performed, and how convincingly it is evaluated under scale. The review argues that the literature already provides strong evidence that flat or weakly structured skill selection becomes unreliable as skill libraries grow, and that structure-aware approaches are promising. However, it also argues that existing work does not yet provide a clear cross-family comparison of which representational properties most effectively distinguish semantically similar but procedurally different skills while remaining retrieval-efficient. This gap motivates the research questions stated at the end of the review.

2.2 Conceptual Dimensions of Skill Retrieval

A useful conceptual distinction is needed before comparing the literature. A *skill artifact* is the actual stored capability, such as a prompt module, executable program, or folder of instructions and scripts. A *selection representation* is the form in which that skill is made available for retrieval or routing, such as a short textual description, a structured schema, a hierarchy position, or a graph node. A *retrieval policy* is the mechanism that selects candidates, such as direct prompt exposure, dense retrieval, hierarchical traversal, graph exploration, or hybrid reranking. Finally, an *evaluation protocol* specifies what success means: retrieval accuracy, downstream task success, context efficiency, robustness under scale, or some combination of these dimensions. This distinction matters because many systems are discussed as if artifact, representation, and retrieval policy were the same thing. In practice, however, a skill may be stored as code or documentation while being represented for selection through only a short textual descriptor. Similarly, a system may use a hierarchical index without changing the internal structure of each skill. Making these layers explicit allows a clearer comparison of what prior work actually contributes.

2.3 Why Skills Are a Distinct Retrieval Problem

The first important question is whether skills are genuinely different from tools for the purposes of retrieval. The recent skill literature strongly suggests that they are. The survey *Agent Skills from the Perspective of Procedural Memory* conceptualizes skills as reusable procedural memories that enable an agent to reuse problem-solving procedures rather than repeatedly reasoning from scratch [6]. This framing is useful because it highlights the procedural nature of skills. The survey organizes the field into acquisition, representation, invocation, and refinement, showing that skills can be encoded as prompts, programs, APIs, or more complex procedural modules. Its main strength is to clarify that a skill is not just a callable function, but a reusable unit of procedure. However, the paper is primarily taxonomical. It does not provide a benchmark or retrieval method that shows how agents should select among many semantically overlapping skills. Its importance for the present thesis lies in problem framing rather than direct methodological evidence.

A similar point is developed more strongly in the *SoK: Agentic Skills — Beyond Tool Use in LLM Agents* [7]. That paper argues that skills extend beyond tools, plans, or memory because they operate as executable procedural modules that can encapsulate complex task knowledge. This is directly relevant to the present topic because it implies that retrieval methods designed for atomic tools may not transfer cleanly to skills. If tools are primarily selected on the basis of function-level relevance, while skills must be selected on the basis of procedural appropriateness, then the retrieval target itself changes. The value of the SoK paper is therefore conceptual: it strengthens the argument that skill retrieval should not be treated as merely a larger version of tool retrieval. However, like the procedural-memory survey, it does not yet solve the practical problem of large-scale skill selection.

Early skill systems further show why this distinction matters. SkillAct introduces textual skill abstractions represented as a tuple consisting of a name, natural-language instructions, and example trajectories, all appended directly to the prompt of a baseline agent [9]. The paper demonstrates that skill abstractions can improve performance, especially by reducing execution errors in ALFWorld. Its contribution is important because it provides an early and transparent baseline for skill-augmented agents. However, its retrieval model is effectively absent: the available skills are exposed as a flat prompt list. This means the system works only because the experimental setting is small enough for full prompt inclusion to remain manageable. The paper therefore supports the claim that skills are useful, but also exposes the weakness of flat prompt-level skill access when libraries become large or internally redundant.

Voyager provides a stronger architectural example of a skill library [10]. It stores successful behaviors as executable code and indexes them by embeddings of their natural-language descriptions. This is significant because it shows that skills can function as reusable external capabilities rather than one-off generated plans. The paper demonstrates that a growing skill library improves exploration and transfer in Minecraft. However, its retrieval mechanism remains shallow: skills are recalled through embedding similarity over descriptions rather than through procedural structure. As a result, Voyager is best understood as a strong early precursor to skill-library

architectures rather than a complete answer to scalable skill retrieval. It demonstrates the utility of externalized skills, but does not study what happens when multiple skills share similar language while differing in operational logic.

2.4 Evidence That Skill Scaling Is a Real Problem

The next major theme in the literature concerns whether the scaling problem is real and, if so, what exactly causes it. Three recent papers are especially important here: *SkillsBench*, *When Single-Agent with Skills Replace Multi-Agent Systems and When They Fail*, and *Agent Skills: A Data-Driven Analysis of Claude Skills* [11, 12, 13]. Together, these papers shift the discussion away from generic claims about capability growth and toward more specific evidence about skill-library behavior under scale.

SkillsBench evaluates whether skills actually help LLM agents across diverse tasks by comparing conditions with no skills, curated skills, and self-generated skills [11]. Its most useful finding is not merely that curated skills improve performance, but that smaller and more focused skill sets often outperform larger and more comprehensive ones. This matters because it suggests that the effectiveness of a skill library depends not only on the presence of reusable procedures, but also on how tightly they are scoped and represented. The benchmark is valuable because it tests skill usefulness across domains rather than assuming that more skills are always better. However, it only weakly addresses the retrieval problem. The relevant curated skills are effectively assumed to be available, so the study does not directly evaluate how the correct skill should be retrieved from a large ambiguous library. Its contribution is therefore diagnostic rather than decisive: it shows that skill packaging matters, but not how scalable skill selection should be performed.

The paper *When Single-Agent with Skills Replace Multi-Agent Systems and When They Fail* is even more directly relevant [12]. It formalizes skills as descriptor-policy-backend tuples and studies how skill selection behaves as the number of available skills increases. Its most important contribution is the finding that performance remains relatively stable until a threshold is reached, after which selection accuracy drops sharply. Crucially, the paper shows that this degradation is driven not merely by the number of skills, but by semantic confusability among them. This is one of the strongest direct pieces of evidence for the present thesis. It isolates semantic overlap as the core failure mode of flat skill selection. At the same time, the paper is weaker as a solution paper. Its proposed hierarchical mitigation is intentionally simple and does not determine which hierarchy design or retrieval structure is best. Nor does it represent skills through rich procedural structure. The paper therefore motivates the research problem very strongly, but does not yet solve it.

The data-driven analysis of Claude skills contributes a different kind of evidence [13]. Rather than proposing a new retrieval method, it measures the properties of a large public skill ecosystem, including redundancy, length variation, and supply-demand imbalance. This is highly relevant because it shows that real skill libraries are already noisy, redundant, and unevenly structured. The paper is particularly useful for context-efficiency arguments, since it shows that while many

skills are short, a non-trivial minority are long enough to create prompt-budget pressure if loaded indiscriminately. It also highlights that discovery often happens through lightweight metadata before full instructions are loaded. However, the paper is not a retrieval-comparison study. Its contribution is ecosystem diagnosis rather than direct evidence about which retrieval methods work best. Still, it supports the claim that scalable discovery and selective loading are practical necessities rather than theoretical luxuries.

2.5 Flat and Weakly Structured Skill Representations

Having established that skills are distinct from tools and that large skill libraries create real confusability problems, the next question is how existing work represents skills for selection. The simplest family of representations remains flat or weakly structured textual skills, as seen in SkillAct and parts of Voyager [9, 10]. These approaches work because they make reusable procedures accessible to the model with minimal infrastructure. Their strengths are simplicity, interpretability, and ease of authoring. Their weakness is that the differentiating information remains embedded in free-form text. A flat description can indicate what a skill is roughly for, but it may not clearly encode conditions under which one apparently similar skill should be preferred over another. This limitation becomes especially serious when the representation available to the retriever is shorter and shallower than the full skill artifact. Flat prompt-based skill systems therefore make a valuable baseline, but not a scalable endpoint.

2.6 Structure-Aware Skill Representations

A more promising direction is to introduce structure explicitly into the skill representation itself. One example is *Learning Agent Skills from Demonstrations and Generating Knowledge Graph*, which decomposes complex demonstrations into meta-skills, generates behavior trees, and transforms the resulting information into a knowledge graph [14]. This paper is relevant because it shows that skills can be represented through explicit internal structure rather than only through free-form descriptions. By moving from trajectories to behavior trees and graph representations, it makes the procedural nature of skills more visible and interpretable. However, the paper is heavily tailored to robotics assembly domains and relies on a predefined meta-skill vocabulary. Moreover, the graph serves more as an organizational and visualization layer than as a thoroughly evaluated large-scale retrieval mechanism. Its main value for the present review is therefore to support the general idea that skills can be decomposed into richer procedural structures.

2.7 Hierarchical Skill Retrieval

Two more directly relevant large-scale skill infrastructure papers are AgentSkillOS and SkillNet [15, 16]. AgentSkillOS organizes skills into a hierarchical capability tree and then uses this

structure for retrieval and DAG-based orchestration. Its strongest contribution is empirical: it demonstrates that structured retrieval can approximate oracle skill selection and that structured orchestration improves downstream task performance compared with flat invocation. This makes it one of the most important system-level papers for the present thesis. However, the capability tree assumption is also its main limitation. Real skills often belong to multiple overlapping functional categories, and a strict tree risks brittle routing decisions. Moreover, the evaluation focuses on overall system performance and orchestration quality rather than isolating retrieval accuracy under highly confusable skill competitors. AgentSkillOS therefore provides strong evidence that structure helps, but weaker evidence about which structural properties are specifically responsible for disambiguating semantically similar skills.

2.8 Graph-Based and Relational Skill Organization

SkillNet takes a broader relational approach [16]. It introduces a taxonomy layer, a relation graph with typed edges such as similarity, composition, and dependency, and a package layer for deployment. This paper is particularly relevant because it refuses to treat skills as isolated items and instead models them as connected assets within a structured network. This is conceptually closer to the needs of large skill ecosystems, where similarity and compositionality often coexist. The main limitation is that the paper functions more as a full infrastructure proposal than as a clean retrieval-comparison study. It provides evidence that structured skill libraries improve downstream performance, but offers less fine-grained analysis of retrieval accuracy, context cost, or robustness under semantic confusability. It therefore strengthens the case for relational structure, but stops short of determining how such structure should be exploited for runtime disambiguation.

Hierarchical retrieval deserves separate attention because it is the clearest structure-aware response to scaling so far. Both *When Single-Agent with Skills Replace Multi-Agent Systems and When They Fail* and AgentSkillOS suggest that coarse-to-fine routing can mitigate the collapse of flat selection under scale [12, 15]. The core intuition is straightforward: forcing the agent to choose directly from a large flat pool is harder than guiding it through a sequence of smaller local decisions. This can reduce both the search space and the prompt burden. However, hierarchy introduces its own assumptions. It assumes that an early routing decision can be made reliably and that the correct skill is reachable through a small number of functional branches. When skills overlap across multiple categories, or when the hierarchy itself is derived from semantics rather than procedure, these assumptions can fail. Thus, hierarchical skill retrieval appears promising as a scalability aid, but it is not yet clear whether it resolves the deeper procedural disambiguation problem or merely postpones it.

Graph-based and relational organization provide a different response to that problem. By allowing multi-parent and typed relationships, graphs potentially model the reality of skill ecosystems more faithfully than strict trees. SkillNet is the clearest direct skill example, while the knowledge-graph paper on learned skills from demonstrations shows that structured relational

representations are at least plausible at the skill level [16, 14]. The main promise of graph-based organization is that it can encode not only similarity but also composition, dependence, and workflow compatibility. However, graph approaches also face a major difficulty: graph quality depends on how nodes and edges are constructed. If similarity edges are derived from the same shallow semantics that already confuse flat retrieval, then the graph may reproduce the original ambiguity rather than solve it. This means that graph structure is not automatically beneficial; its utility depends on whether the encoded relations capture meaningful procedural distinctions.

2.9 Transferable Ideas from Tool and Document Retrieval

At this point it is reasonable to ask whether ideas from the tool-selection and document-retrieval literature can help. They can, but only as *transferable methodological evidence*, not as direct proof about skill retrieval. For example, SciToolAgent shows that graph-based organization of tools can improve planning and multi-step execution by representing input-output formats, dependencies, and safety relations explicitly [17]. This paper is useful because it demonstrates the value of dependency-aware structured retrieval. However, its unit of organization is the tool, not the skill, and its graph is manually curated in a highly specialized domain. Similarly, LATTICE demonstrates that hierarchical traversal with calibrated path scoring can outperform flat retrieve-then-rerank pipelines for document retrieval [18]. This is methodologically interesting because it shows that LLMs can be used to navigate an index rather than only rerank retrieved candidates. Yet its direct transfer to skills is limited because documents are not procedural capability units. ToolLLM and Gorilla further establish the practical importance of retrieval when capability spaces become large, but they remain fundamentally tool-centric and do not address procedural skill structure [5, 4]. These papers should therefore be used cautiously: they provide ideas for hierarchy, graph structure, reranking, and representation design, but not direct evidence that those methods solve skill retrieval.

2.10 Evaluation Practices and Unresolved Gaps

One of the clearest gaps in the literature emerges when evaluation practices are examined closely. Many papers report overall task success, but fewer isolate retrieval quality itself. SkillsBench measures whether skills help, but not large-library disambiguation [11]. AgentSkillOS and SkillNet demonstrate strong system-level gains, but retrieval, orchestration, and planning improvements are partly entangled [15, 16]. The Claude-skills analysis reveals redundancy and selective-loading pressure, but does not compare retrieval methods experimentally [13]. The Li paper on single-agent skill failure directly targets semantic confusability, which makes it especially important, yet its mitigation strategy is intentionally simple [12]. Across these works, three recurring evaluation limitations appear. First, semantically confusable but procedurally distinct skills are not always tested explicitly. Second, context efficiency is often discussed qualitatively rather than measured directly in tokens, shortlist size, or cost. Third, some

structured approaches are compared against relatively weak baselines, making it difficult to isolate how much value comes from structure-aware selection itself rather than broader system engineering improvements.

These limitations matter because the present research problem is more specific than generic retrieval quality. A method may retrieve useful skills on average and still fail on the exact cases that matter most for a personal skill library: cases where surface similarity is high but the correct procedure is different. Likewise, a method may improve overall task performance through better orchestration without demonstrating that the representation itself improves retrieval under confusability. The strongest literature therefore points in a common direction without fully closing the question. Skills are useful. Large skill libraries create real ambiguity. Structure-aware organization appears promising. Yet the literature still does not clearly determine which representation best preserves procedural distinguishability while remaining efficient and effective at runtime.

2.11 Synthesis and Research Questions

The literature therefore suggests that flat skill selection becomes unreliable as skill libraries grow and semantic confusability increases. Although structure-aware approaches are promising, it remains unclear which representational properties most effectively preserve procedural distinguishability while remaining context-efficient and useful in downstream execution. This gap motivates the following research questions:

RQ1: How can agent skills be represented and retrieved so that agents can accurately select among semantically similar but procedurally distinct skills as skill libraries scale?

RQ2: To what extent do structure-aware skill representations improve retrieval accuracy, context efficiency, and downstream task performance compared with flat description-based skill retrieval?

These questions arise naturally from the literature rather than being imposed on it. Existing work has already shown that flat selection becomes unreliable under scale, that the problem is linked to semantic confusability, and that hierarchical or graph-based organization may help. What remains unresolved is which representational properties most effectively preserve procedural differences between competing skills, and how that effect should be measured. The remainder of the project is therefore best understood not as a generic study of agent scaling, but as a focused investigation into structure-aware skill retrieval under semantic confusability.

2.12 Update Notes for Thesis Revision

Placeholder. Later revision: update this chapter to align the final synthesis with the newer Chapter 1 research questions. The content above should remain the base literature review, but the final thesis should integrate newer papers such as SkillRouter, SkillRet, and Skill Retrieval Augmentation more directly and revise the final RQ wording.

Chapter 3

Conceptual Framework

3.1 Skill Artifacts

A skill artifact is the stored capability unit. In this thesis, the canonical artifact is a skill directory containing a `SKILL.md` file and optional supporting files such as scripts, examples, reference documents, or templates. The artifact is the complete procedural object available to the system.

The artifact is not necessarily what the retriever sees. A complete skill may be too long to expose in every retrieval decision, especially when the library contains hundreds or thousands of skills. This motivates a distinction between the stored artifact and the selection representation.

3.2 Selection Representations

A selection representation is the version of a skill used for retrieval or routing. It may be a short card containing name and description, a full `SKILL.md` document, a structured schema, a hierarchy node, or a graph node with typed edges.

The key question is what information this representation preserves. For the thesis problem, the most important candidate fields are:

- skill name and family;
- description and use cases;
- required inputs and source type;
- expected outputs and artifact shape;
- preconditions and assumptions;
- workflow steps and ordering constraints;

- dependencies, resources, scripts, and external services;
- negative usage boundaries, such as `not_for` conditions;
- meta-level task role, such as editing a skill versus using a skill.

3.3 Retrieval Policies

A retrieval policy is the mechanism that selects candidate skills from the representation. The policy may be:

- direct main-agent selection from visible skill cards;
- lexical retrieval using BM25 or TF-IDF;
- dense embedding retrieval;
- structured field scoring;
- neural reranking;
- deterministic schema reranking;
- hierarchical or tree routing;
- graph traversal or graph-based expansion;
- a hybrid pipeline combining candidate generation and reranking.

This separation is essential. A method can fail because the representation lacks the needed information, or because the retrieval policy cannot use information that is present. Conversely, a strong retriever may still fail if the representation compresses away procedural boundaries.

3.4 Semantic Confusability and Procedural Distinctness

This thesis defines semantic confusability as a condition where multiple skills appear close at the level of name, description, domain, or trigger phrases. It defines procedural distinctness as a condition where those same skills require materially different execution procedures, inputs, outputs, constraints, or verification criteria.

The target failure mode occurs when a retrieved skill is semantically plausible but procedurally inappropriate. This is why the benchmark does not simply evaluate whether the correct broad domain is retrieved. The system must identify the correct procedure within a set of plausible neighbours.

3.5 Working Hypotheses

The experiments are guided by four working hypotheses:

1. Flat metadata will degrade under scale because it under-specifies procedural distinctions.
2. Full skill artifacts will improve over short descriptions because they expose more procedural information, but may increase context and computation cost.
3. Structured procedural fields will improve disambiguation when the user request contains enough evidence to match those fields.
4. Hybrid retrieval-reranking methods will perform best when dense retrieval provides broad candidate coverage and procedural reranking resolves near-neighbour confusion.

Placeholder. Add a diagram showing the three-layer framework: artifact -> selection representation -> retrieval policy -> candidate skills -> main agent.

Chapter 4

Benchmark Design

4.1 Design Goals

The benchmark is designed to test skill retrieval under the failure mode that motivates the thesis: semantic confusability between procedurally distinct skills. It is not enough for the benchmark to contain many unrelated skills. The benchmark must contain clusters where several skills sound relevant to the same request, but only one skill is procedurally correct.

The benchmark therefore has three goals:

1. create controlled near-neighbour skill clusters;
2. scale the library with background skills so retrieval is not performed in a toy setting;
3. validate that gold labels remain defensible even when non-core background skills are considered.

4.2 Skill Library

The current benchmark contains 1006 skills:

- 67 controlled core skills;
- 920 generated background-scale skills;
- 15 public imported background skills;
- 4 support or email skills.

The controlled core skills are the main evaluation targets. They are organized into task families such as document processing, code and GitHub workflows, planning and meetings, reading and

research, security and privacy, metrics and observability, browser or web automation, reply and messaging, data and spreadsheets, and skill lifecycle operations.

Placeholder. Add a table listing all controlled clusters, each cluster’s skills, and the procedural distinction being tested.

4.3 Confusable Clusters

A valid confusable cluster must satisfy three conditions. First, the skills must overlap in surface meaning, name, description, or trigger phrases. Second, the skills must differ in operationally important procedure. Third, choosing the wrong skill must matter for the resulting workflow or artifact.

For example, a document cluster may include document summarisation, field extraction, citation grounding, document normalisation, and layout-preserving conversion. These skills all involve documents, but they differ in input requirements, output shape, workflow steps, and verification criteria.

Placeholder. Insert 3-5 detailed cluster examples. For each example, include the prompt, gold skill, near-neighbour distractors, and the exact procedural distinction.

4.4 Background Scale Skills

Background skills are included to make the library scale more realistic. They are not the main research target, but they test whether retrieval methods remain robust when many plausible but non-gold candidates are present. Background skills also help reveal whether a supposedly controlled prompt accidentally matches a non-core skill better than the intended gold skill.

The benchmark includes generated background skills and a small number of public imported skills. Generated skills are useful for controlled scale, while public skills improve ecological validity by introducing less curated wording and real-world skill variation.

4.5 Gold Labels and Acceptable Alternatives

Each evaluated prompt has a gold-label intended skill. A gold label is valid when a careful human reading shows that the prompt’s procedural requirements align more strongly with the gold skill than with its near neighbours.

Some cases may require acceptable alternatives. This is especially true when a background skill is domain-specific and could reasonably satisfy the prompt even if it was not the original target. Acceptable alternatives should be recorded explicitly rather than silently counted as failures or successes.

Placeholder. Add the final acceptable-alternatives policy and include examples from the document field extraction and reply drafting cases.

4.6 Validation Rubric

The benchmark is validated through a multi-step rubric:

1. Integrity: all prompts and gold skills resolve.
2. Procedural alignment: each prompt contains requirements that justify the gold skill.
3. Semantic confusability: each prompt has plausible near-neighbour skills.
4. Prompt leakage: prompts should not reveal exact skill names in a way that makes retrieval trivial.
5. Scale regime: the library should be large enough to test scalable retrieval pressure.
6. Selector pressure: baseline methods should show meaningful performance differences.
7. Progressive-disclosure baseline: the normal main-agent selection pattern should be evaluated where feasible.
8. Downstream validation: retrieval differences should be tested for downstream task impact.

Placeholder. Move the detailed pass/fail thresholds from the benchmark methodology rubric into an appendix table.

Chapter 5

Methodology

5.1 Experimental Overview

The methodology is a controlled comparison of representation-and-retrieval conditions. Each method receives the same user prompts and skill library, produces a ranked list of candidate skills, and is evaluated against gold labels and acceptable alternatives.

The independent variables are the selection representation and retrieval policy. The main dependent measures are top-1 accuracy, top-k recall, mean reciprocal rank, context cost, latency, and qualitative failure type.

5.2 Representation Conditions

The current experiments include the following representation conditions:

- **R1 flat metadata:** skill name, family, tags, and short description.
- **Full artifact text:** the full `SKILL.md` text.
- **R2 structured procedural card:** explicit procedural fields such as use cases, preconditions, workflow, outputs, and negative boundaries.
- **R3 dependency/resource-aware representation:** structured procedural fields plus dependency and resource signals.
- **R4 graph edges:** explicit relation files for future graph experiments.

Placeholder. Clarify which R-levels are used in the final experiments. If R3 and R4 are not central, explain that they are prepared but not fully evaluated.
--

5.3 Retrieval Strategies

The current method families are:

- **M0 progressive disclosure:** the main agent sees visible skill cards and decides which full skill documents to load.
- **M1 lexical flat retrieval:** BM25 or TF-IDF over flat metadata.
- **M2 embedding retrieval:** dense retrieval over descriptions or full skill artifacts.
- **M3 schema-based retrieval:** structured scoring over procedural fields.
- **M6 local hybrid retrieval:** first-stage retrieval followed by deterministic schema reranking.
- **M7 provider embedding retrieval:** API-based dense retrieval using Qwen embeddings.
- **M8 provider neural reranking:** Qwen embedding retrieval followed by Qwen reranking.
- **M8b provider retrieval plus local schema reranking:** Qwen first-stage retrieval followed by deterministic procedural reranking.

Tree-based and graph-based strategies are currently planned but not yet fully implemented. They remain relevant to the thesis design, but the current strongest evidence comes from full-artifact embedding retrieval combined with procedural schema reranking.

5.4 Schema Reranking

The local schema reranker is deterministic. It scores overlap between the user request and structured skill fields, with stronger weights assigned to fields that should reflect procedural suitability. These include use cases, outputs, preconditions, workflow steps, dependencies, resources, and negative boundaries. The final reranked score combines the first-stage retrieval score with the schema score.

This method is intentionally simple. Its purpose is not to claim that hand-weighted overlap is the best possible architecture, but to test whether explicit procedural fields contain information that helps resolve semantically confusable retrieval cases.

Placeholder. Insert the final exact weighting formula and justify each field weight. Include an ablation if time permits.

5.5 Evaluation Metrics

The retrieval metrics are:

- **Top-1 accuracy:** whether the top-ranked skill is the gold or acceptable skill.
- **Top-k recall:** whether the gold or acceptable skill appears within the top-k candidate set.
- **Mean reciprocal rank:** how highly the gold skill is ranked.
- **Non-core top-1 rate:** how often a background skill outranks the intended controlled skill.

The efficiency metrics are:

- candidate budget;
- representation length;
- token or context cost;
- latency;
- API calls and cache behaviour.

The qualitative evaluation records failure modes such as first-stage exclusion, reranker misweighting, negation failure, object/meta-skill confusion, underspecified prompts, and acceptable-alternative ambiguity.

5.6 Downstream Validation Plan

Downstream validation will test whether retrieval improvements translate into better task outcomes. A smaller subset of prompts will be selected, and the agent will receive either the top-1 selected skill or a top-k candidate set. Generated artifacts will then be graded against task-specific success criteria.

Placeholder. Step 8 is not yet executed. Add final downstream task selection, grading rubric, evaluator design, and results once the experiment is complete.

5.7 Ethics, Copyright, and Reproducibility

This project does not currently involve human participants or sensitive personal data. The benchmark consists of synthetic prompts, controlled skill artifacts, generated background skills, and a small number of public skill examples used for research comparison. Any public skill material should be recorded with source information and license notes where available.

Reproducibility is supported by storing benchmark prompts, skill artifacts, representation exports, selector scripts, cached provider outputs, and evaluation reports in the project repository.

Provider-based experiments should report model names, API providers, candidate budgets, cache status, and approximate cost or latency where possible.

Placeholder. Confirm the final ethics status with supervisor guidance. Add any required USYD academic honesty, copyright, AI-use, and data-management statements.

Chapter 6

Results

6.1 Benchmark Validation Results

The current benchmark contains 1006 skills and 67 evaluated prompts. The validation rubric currently shows:

- Step 1 integrity passed: all prompts and gold skills resolve.
- Step 2 procedural alignment passed: all prompts pass field audit, with a small number of watch items under stricter alignment.
- Step 3 semantic confusability passed: most prompts have measurable near-neighbour semantic competition.
- Step 4 prompt leakage passed: no critical exact-name leaks or high-risk leaks.
- Step 5 scale regime passed: the library is large enough to test scalable retrieval pressure.

Placeholder. Insert the final benchmark validation table from the rubric output. Include exact counts and thresholds.

6.2 Local Selector Results

Table 6.1 summarises the current local full-library results.

These results suggest that description-only embedding is fragile under scale, while full-skill retrieval and schema-aware reranking improve performance. The best local method is the hybrid full-skill embedding plus schema reranking condition.

Method	Top-1	Top-5	MRR
M1 BM25 flat	64.2%	86.6%	0.733
M1 TF-IDF flat	58.2%	85.1%	0.697
M2a MiniLM description	47.8%	74.6%	0.603
M2b MiniLM full skill	61.2%	82.1%	0.717
M3 TF-IDF schema	71.6%	95.5%	0.808
M6 BM25 to schema rerank	71.6%	92.5%	0.815
M6 TF-IDF to schema rerank	70.2%	85.1%	0.778
M6 MiniLM full to schema rerank	80.6%	91.0%	0.856

Table 6.1: Current local selector results on the 1006-skill library.

6.3 Provider Results

Qwen provider experiments test whether the benchmark remains difficult under stronger embedding and reranking models. Current full-library results show that Qwen reranking improves over Qwen embedding-only for each representation, but generic neural reranking does not fully solve the benchmark.

Method	Top-1	Top-5	MRR
Qwen R1 embedding only	35.8%	56.7%	0.449
Qwen R1 embedding + rerank	64.2%	67.2%	0.649
Qwen full embedding only	46.3%	71.6%	0.574
Qwen full embedding + rerank	61.2%	65.7%	0.644
Qwen R2 embedding only	47.8%	65.7%	0.563
Qwen R2 embedding + rerank	65.7%	71.6%	0.688

Table 6.2: Current Qwen provider results on the 1006-skill library.

6.4 Hybrid Qwen Retrieval plus Schema Reranking

The strongest current method uses Qwen full-skill embedding as a first-stage candidate generator and the deterministic local schema reranker for final ranking. Table 6.3 shows the effect of candidate budget.

Candidate budget	Top-1	Top-5	MRR
Top-20	68.7%	74.6%	0.715
Top-50	80.6%	88.1%	0.834
Top-100	85.1%	91.0%	0.878

Table 6.3: Qwen full-skill retrieval followed by local schema reranking.

This result supports the main thesis framing: dense retrieval is useful for broad candidate generation, but final selection benefits from explicit procedural information.

6.5 Low-Information Stress Test

A separate low-information prompt set was used to check whether schema reranking depends on explicit procedural cues. Performance drops sharply across methods. The best strict top-1 score is currently 41.7% for MiniLM full-skill embedding. Qwen full-skill retrieval plus local schema reranking reaches 25.0% top-1 and 66.7% top-5.

This suggests that schema reranking is not secretly inferring hidden intent. It works best when the request contains enough procedural evidence to match structured fields. Very underspecified requests may require a query-understanding or clarification step before retrieval.

6.6 Failure Mode Analysis

The strongest current method has 10 strict top-1 failures out of 67 prompts. Four failures are first-stage exclusions where Qwen does not place the gold skill in the top-100. Six failures are reranker, boundary, negation, meta-skill, or annotation issues.

The main failure types are:

- first-stage embedding confuses task object with required procedure;
- schema reranking overweights secondary terms;
- negative usage boundaries are too weak;
- meta-skill tasks such as editing or evaluating a skill are confused with using the named skill;
- some domain-specific competitors may require acceptable-alternative adjudication;
- underspecified prompts create genuine ambiguity.

Placeholder. Add the final failure-mode table with prompt IDs, gold skill, predicted skill, first-stage rank, final rank, and failure category.

6.7 Efficiency Results

Placeholder. Not yet complete. Add candidate-budget latency, API cost, token cost, cache behavior, and top-20/top-50/top-100 trade-off analysis.

6.8 Downstream Results

Placeholder. Not yet complete. Add Step 8 downstream artifact generation results and grading. This should connect retrieval accuracy to actual task success.

Chapter 7

Discussion

7.1 What the Results Suggest about Representation

The current results suggest that the central issue is not simply choosing a stronger retriever. Stronger embeddings and neural reranking improve some conditions, but they do not eliminate semantic confusion. The strongest current result comes from combining broad semantic candidate generation with a reranker that uses explicit procedural fields.

This supports the claim that retrieval strategies and representation quality are coupled. A dense retriever can find plausible candidates quickly, but it may not reliably distinguish skills whose descriptions overlap. A procedural schema can help distinguish candidates, but only when the correct skill appears in the candidate pool and the user request contains enough procedural evidence.

7.2 Information that Appears Important

The benchmark and failure analysis indicate that the following information types are especially important:

- input source type;
- expected output shape;
- workflow steps and ordering;
- preconditions and assumptions;
- resource or dependency requirements;
- negative boundaries;
- whether the task is about using a skill or modifying a skill;

- primary user intent versus secondary content terms.

Placeholder. After final ablations, rank these information types by evidence strength.
--

7.3 Why Architecture Alone Is Not the Thesis

The experiments compare architectures, but the thesis contribution is not simply that one architecture wins. The more important finding is which representational information each architecture preserves or loses. A flat metadata system loses many procedural distinctions. Full-skill embedding exposes more information but may still score based on broad semantic similarity. Structured procedural fields make the relevant distinctions explicit, but depend on authoring quality and request specificity. Hybrid methods work because they combine complementary strengths.

7.4 Failure Modes and Limits

The low-information stress test shows a key limitation: retrieval cannot always solve under-specification. If a request lacks procedural evidence, the correct response may be to ask a clarification question rather than force a skill selection. This means future systems may need a query-understanding stage before retrieval, especially for vague user requests.

Another limitation is that deterministic schema reranking may overfit to the benchmark’s structured fields if not carefully validated. The use of background skills, non-core procedural review, low-information prompts, and provider-based retrieval helps reduce this risk, but further external validation would strengthen the thesis.

7.5 Implications for Agent Skill Systems

The practical implication is that large skill libraries should not rely only on skill names and short descriptions. They should preserve procedural fields that support disambiguation, and retrieval pipelines should separate candidate generation from final procedural selection. The results also suggest that full skill artifacts and structured summaries may both be useful, but for different stages of retrieval.

Placeholder. Add final design recommendations for future agent skill systems, based on final experiments.

Chapter 8

Conclusion

This thesis investigates skill representation for scalable agent skill retrieval. The motivating problem is semantic confusion: as skill libraries grow, many skills can sound similar while requiring different procedures. In such cases, the correct skill may exist, but the retrieval system may surface a plausible yet procedurally inappropriate alternative.

The thesis reframes skill selection as a layered problem involving skill artifacts, selection representations, and retrieval policies. It argues that the key research issue is not only which retriever performs best, but what information must be preserved so that retrieval can distinguish semantically similar but procedurally distinct skills.

Current results support the value of explicit procedural information. Flat metadata and description-only embedding retrieval are fragile under scale. Full skill artifacts help, and structured procedural representations help further when the request contains enough evidence. The strongest current result combines dense retrieval over full skill artifacts with structure-aware schema reranking, suggesting that semantic retrieval and procedural reranking are complementary.

Placeholder. Rewrite this conclusion after final experiments. Include final answers to RQ1 and RQ2, final limitations, and future work.

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Appendix A

Benchmark Rubric

Placeholder. Paste or summarize the final benchmark methodology rubric from `skill_benchmark/notes/benchmark_methodology_rubric.md`. Include pass thresholds for Steps 1-8.

Appendix B

Prompt and Skill Examples

Placeholder. Add representative prompt examples from each controlled cluster. For each example, show the user request, gold skill, plausible distractors, and why the gold label is procedurally correct.